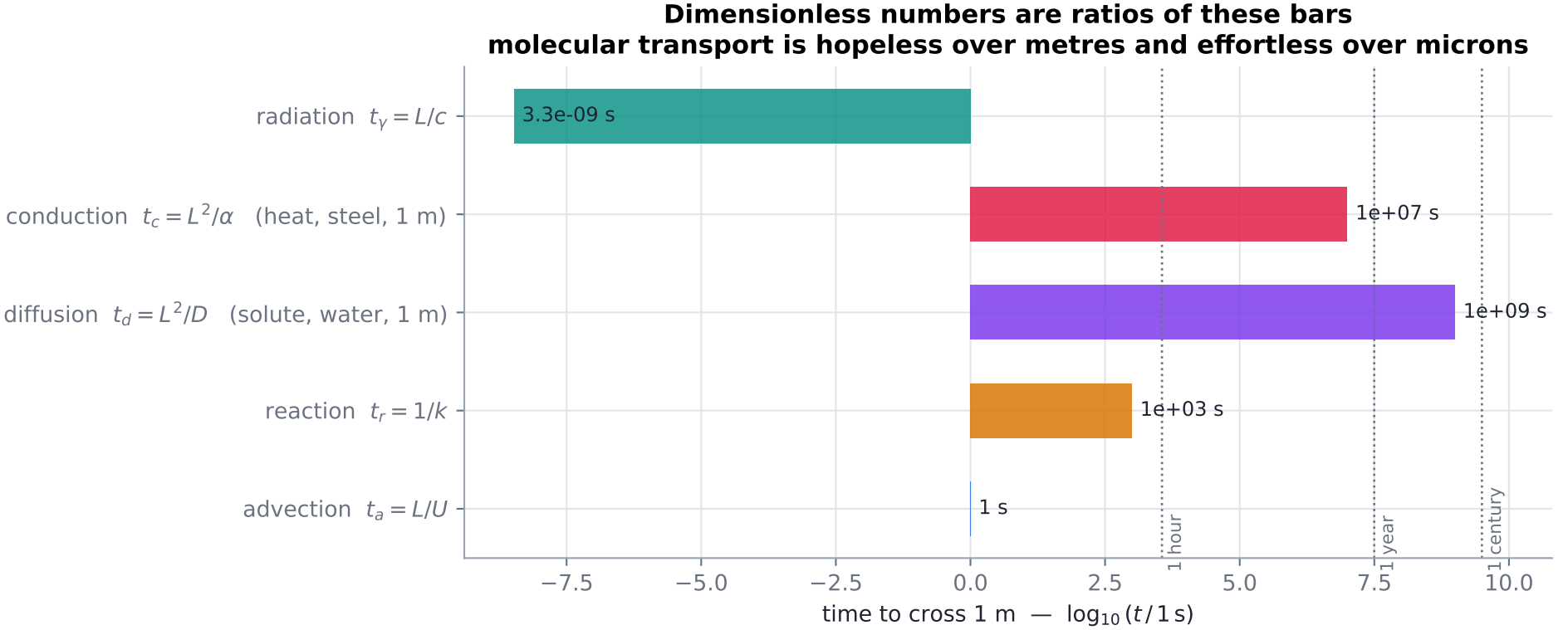
Dimensionless numbers are ratios of these bars
molecular transport is hopeless over metres and effortless over microns
radiation $t_\gamma = L/c$
conduction $t_c = L^2/\alpha$ (heat, steel, 1 m)
diffusion $t_d = L^2/D$ (solute, water, 1 m)
reaction $t_r = 1/k$
advection $t_a = L/U$
3.3e-09 s
1e+07 s
1e+09 s
1e+03 s
1 s
1 hour
1 year
1 century
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
time to cross 1 m — $\log_{10}(t/1\,\mathrm{s})$